

## ECE 341 Homework 2

The problem numbers listed below are from the Lathi textbook, 3<sup>rd</sup> edition.

| Prob. Number     | Course Outcome | Points |
|------------------|----------------|--------|
| 2.2-1<br>(b, d)  | 1.b            | 20     |
| 2.3-2            | 1.b            | 20     |
| 2.4-18 *         | 1.b            | 10     |
| 2.4-27<br>(b, f) | 1.b            | 30     |
| 2.5-1            | 7.c            | 20     |
| TOTAL:           |                | 100    |

**\* Hint: You can use Table 2.1 of the textbook (Page 176) to find the convolution.**

**1-b.** Evaluate and classify signals to identify unstable systems.

**7-c.** Apply new knowledge to understand continuous and discrete systems.

2.2.1 (b) Given the system described by

$$\ddot{y}(t) + 2\dot{y}(t) + 5y(t) = \ddot{x}(t) - 5x(t),$$

with initial conditions

$$\begin{aligned} y_{\text{zir}}(0) &= 4 \\ \dot{y}_{\text{zir}}(0) &= -1, \end{aligned}$$

begin by setting  $P(D)x(t)$  to zero to find the zero-input response, defined as  $y_{\text{zir}}(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t}$ . To find  $\lambda_1$  and  $\lambda_2$ , we will solve the characteristic equation  $\lambda^2 + 2\lambda + 5 = 0$ , which produces two complex roots,

$$\lambda_1 = -1 + j2, \quad \lambda_2 = -1 - j2.$$

We can pick one of two ways to solve for  $c_1$  and  $c_2$ : we can keep the ZIR in complex form and solve for the  $c_1$  and  $c_2$ , or we can solve for the constants  $c_1$  and  $c_2$  more easily by converting the ZIR to a real form. In that case,  $c_1$  and  $c_2$  must be conjugates, expressed as

$$\begin{aligned} c_1 &= \frac{c}{2} e^{j\theta} \\ c_2 &= \frac{c}{2} e^{-j\theta}. \end{aligned}$$

Substituting these new constants into our expression for  $y_{\text{zir}}$ ,

$$y_{\text{zir}}(t) = \frac{c}{2} e^{j\theta} e^{(-1+j2)t} + \frac{c}{2} e^{-j\theta} e^{(-1-j2)t} = \frac{c}{2} [e^{-t} e^{j(2t+\theta)} + e^{-t} e^{j(-2t-\theta)}].$$

Simplifying using Euler's formula,  $e^{jx} = \cos(x) + j \sin(x)$ ,

$$y_{\text{zir}}(t) = \frac{ce^{-t}}{2} [\cos(2t + \theta) + j \sin(2t + \theta) + \cos(-2t - \theta) + j \sin(-2t - \theta)].$$

Since  $\cos(-x) = \cos(x)$  and  $\sin(-x) = -\sin(x)$ , we may further reduce the ZIR to

$$\frac{ce^{-t}}{2} (2 \cos(2t + \theta)) = ce^{-t} \cos(2t + \theta).$$

Now, we may easily solve for  $c$  and  $\theta$  as real values by setting  $t = 0$  for  $y_{\text{zir}}(t)$  and  $\dot{y}_{\text{zir}}(t)$ ,

$$\begin{aligned} y_{\text{zir}}(t) &= ce^{-t} \cos(2t + \theta) \\ \dot{y}_{\text{zir}}(t) &= -ce^{-t} \cos(2t + \theta) - 2ce^{-t} \sin(2t + \theta). \end{aligned}$$

Setting these expressions equal to the given initial conditions at  $t = 0$ ,

$$\begin{aligned} y_{\text{zir}}(0) &= c \cos(\theta) = 4 \\ \dot{y}_{\text{zir}}(0) &= -c \cos(\theta) - 2c \sin(\theta) = -1. \end{aligned}$$

Solving for  $c \cos(\theta)$  and  $c \sin(\theta)$  using a system of equations,

$$\begin{bmatrix} 1 & 0 \\ -1 & -2 \end{bmatrix} \begin{bmatrix} c \cos(\theta) \\ c \sin(\theta) \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 4 \\ -1 & -2 & -1 \end{bmatrix} \Rightarrow \begin{aligned} c \cos(\theta) &= 4 \\ c \sin(\theta) &= -1.5 \end{aligned}$$

Considering both terms to be the coordinates of a complex number mapped onto the x-y plane, we may solve for  $c$  and  $\theta$  as  $c = \sqrt{4^2 + 1.5^2} = 4.272$  and  $\theta = \tan^{-1} \frac{-1.5}{4} = -0.359$ , we can substitute this back into our solution for  $y_{\text{zir}}(t)$ , which gives us the complete ZIR,

$$y_{\text{zir}}(t) = 4.272e^{-t} \cos(2t - 0.359).$$

Finally, we will convert our values for  $c$  and  $\theta$  back to  $c_1$  and  $c_2$  if we want, as

$$c_1 = \frac{c}{2}e^{j\theta} = 2.136e^{-j0.359}$$

$$c_2 = \frac{c}{2}e^{-j\theta} = 2.136e^{j0.359}.$$

The result is a decaying sinusoid, which is to be expected given complex-valued characteristic roots.

See Fig. 1 for a plot of the ZIR.

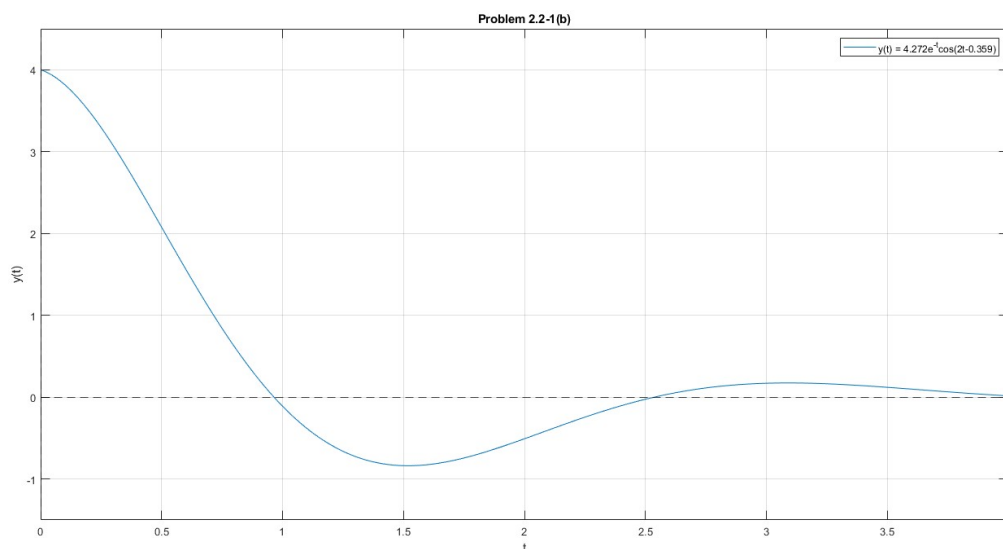


Figure 1: ZIR response

- (d) Given the system described by  $D^2 + 2D + 10\{y(t)\} = (D^5 - D)\{x(t)\}$  with initial conditions  $y_{\text{zir}}(0) = \dot{y}_{\text{zir}}(0) = 1$ , the characteristic equation  $\lambda^2 + 2\lambda + 10 = 0$ , produces solutions

$$\lambda_1 = -1 + j3$$

$$\lambda_2 = -1 - j3.$$

Using the same process as for the prior problem, the corresponding ZIR and its derivative are given by

$$y_{\text{zir}}(t) = ce^{-t} \cos(3t + \theta)$$

$$\dot{y}_{\text{zir}}(t) = -ce^{-t} \cos(3t + \theta) - 3ce^{-t} \sin(3t + \theta).$$

Solving for  $c \cos(\theta)$  and  $c \sin(\theta)$ ,

$$c \cos(\theta) = 1$$

$$c \sin(\theta) = -\frac{2}{3}.$$

Therefore,  $c = 1.202$  and  $\theta = -0.588$ , so

$$c_1 = 0.6e^{-j0.588}$$

$$c_2 = 0.6e^{j0.588},$$

and the resultant ZIR is

$$y_{\text{zir}}(t) = 1.202e^{-t} \cos(3t - 0.588).$$

2.3.2 Given the system described by  $(D^2 + 4D + 3)y(t) = (D + 5)x(t)$ , we may find the impulse response  $h(t)$  by setting the input  $x(t) = \delta(t)$ , so that

$$(D^2 + 4D + 3)h(t) = (D + 5)\delta(t).$$

Because  $M < N$ , we may assume that there is no delta component in the impulse response, and it is exclusively composed of the system's characteristic modes. Solving for  $\lambda_1$  and  $\lambda_2$ ,

$$\begin{aligned}\lambda_1 &= -3 \\ \lambda_2 &= -1,\end{aligned}$$

and so

$$h(t) = (c_1 e^{-3t} + c_2 e^{-t})u(t).$$

Solving for constants  $c_1$  and  $c_2$ , we'll set "placeholder" initial jumps of  $h(0^+) = K_1$  and  $\dot{h}(0^+) = K_2$ . Then, taking derivatives and limiting ourselves to the system response at  $t = 0$  (meaning we can ignore its characteristic modes for the time being),

$$\begin{aligned}\dot{h}(t) &= K_1 \delta(t) \\ \ddot{h}(t) &= K_1 \dot{\delta}(t) + K_2 \delta(t).\end{aligned}$$

Then, taking the system description for  $t = 0$  and replacing  $\dot{h}(0)$  and  $\ddot{h}(0)$  with the aforementioned impulse terms,

$$\begin{aligned}(K_1 \dot{\delta}(t) + K_2 \delta(t)) + 4K_1 \delta(t) + h(0) &= \dot{\delta}(t) + 5\delta(t), \\ K_1 \dot{\delta}(t) + (4K_1 + K_2)\delta(t) + h(0) &= \dot{\delta}(t) + 5\delta(t).\end{aligned}$$

Matching coefficients of  $\dot{\delta}(t)$  and  $\delta(t)$  on either side,

$$\begin{aligned}K_1 &= 1 \\ 4K_1 + K_2 &= 5,\end{aligned}$$

and so, solving for  $K_1$  and  $K_2$ ,

$$\begin{aligned}K_1 &= 1 \\ K_2 &= 1.\end{aligned}$$

Next, finding constants  $c_1$  and  $c_2$  using  $h(t) = (c_1 e^{-3t} + c_2 e^{-t})u(t)$  and  $\dot{h}(t) = (-3c_1 e^{-3t} - c_2 e^{-t})u(t) + (c_1 e^{-3t} + c_2 e^{-t})\delta(t)$ ,

$$\begin{aligned}h(0^+) &= c_1 + c_2 = 1 \\ \dot{h}(0^+) &= -3c_1 - c_2 = 1,\end{aligned}$$

so that  $c_1 = -1$  and  $c_2 = 2$ .

Therefore, the total impulse response is given as

$$h(t) = (-e^{-3t} + 2e^{-t})u(t).$$

Using the simplified impulse matching method to confirm the results,

$$h(t) = b_0 \delta(t) + [P(D)y_n(t)]u(t), \tag{1}$$

assume  $b_0 = 0$ ,  $P(D) = D + 5$  and solve for  $y_n(t)$ , assuming  $y_n(0) = 0$  and  $\dot{y}_n(0) = 1$  as

$$\begin{aligned} h(t) &= c_1 e^{-3t} + c_2 e^{-t} \\ \dot{h}(t) &= -3c_1 e^{-3t} - c_2 e^{-t} \end{aligned}$$

$$\begin{aligned} h(0) &= c_1 + c_2 = 0 \\ \dot{h}(0) &= -3c_1 - c_2 = 1 \end{aligned}$$

$$c_1 = -0.5, \quad c_2 = 0.5$$

$$y_n(t) = -0.5e^{-3t} + 0.5e^{-t}.$$

Solving for  $h(t)$ ,

$$h(t) = [(D + 5)(-0.5e^{-3t} + 0.5e^{-t})]u(t) = [(1.5e^{-3t} - 0.5e^{-t}) + (-2.5e^{-3t} + 2.5e^{-t})]u(t).$$

Simplyfing,

$$h(t) = (-e^{-3t} + 2e^{-t})u(t),$$

which matches the initial solution for  $h(t)$ .

2.4.18 Given the system whose unit impulse response is  $h(t) = [(1 - 2t)e^{-2t}]u(t)$ , we can find the ZSR of the system,  $y(t)$ , as the convolution of the input  $x(t)$  and the impulse response,

$$h(t) * x(t) = \int_0^\infty h(\tau)x(t - \tau)d\tau,$$

or

$$x(t) * h(t) = \int_0^\infty x(\tau)h(t - \tau)d\tau.$$

For input  $x(t) = u(t)$ , the convolution of  $[(1 - 2t)e^{-2t}]u(t) * u(t)$  simplifies via the distributive property as

$$e^{-2t}u(t) * u(t) - 2te^{-2t}u(t) * u(t).$$

The two resulting convolutions are most closely matched in Table 2.1 as

$$\begin{aligned} te^{\lambda_1 t}u(t) * e^{\lambda_2 t}u(t) &= \frac{e^{\lambda_2 t} - e^{\lambda_1 t} + (\lambda_1 - \lambda_2)te^{\lambda_1 t}}{(\lambda_1 - \lambda_2)^2}u(t) \\ e^{\lambda t}u(t) * u(t) &= \frac{1 - e^{\lambda t}}{-\lambda}u(t). \end{aligned}$$

Solving for the first convolution using  $\lambda = -2$ ,

$$e^{-2t}u(t) * u(t) = \frac{1 - e^{-2t}}{-2}u(t).$$

Solving for the second, using  $\lambda_1 = -2$  and  $\lambda_2 = 0$ ,

$$2te^{-2t}u(t) * u(t) = 2\frac{-e^{-2t} - 2te^{-2t}}{4}u(t).$$

Therefore,

$$y_{\text{zsr}}(t) = [-\frac{1}{2}e^{-2t} - \frac{1}{2}(-e^{-2t} - 2te^{-2t})]u(t) = te^{-2t}.$$

- 2.4.27 b) For  $c(t) = A[u(t-3) - u(t-5)] * B[u(t+5) - u(t+3)]$ , we can define areas for no overlap where  $t < -2$  and  $t > 2$ , partial overlap where  $-2 < t < 0$  and  $0 < t < 2$ , and full overlap at  $t = 0$ . Assuming  $B = 2A$ , arbitrarily, for the sake of the example, Fig. 2 shows the defined areas as well as the graph of the convolution itself as a function of  $t$ .

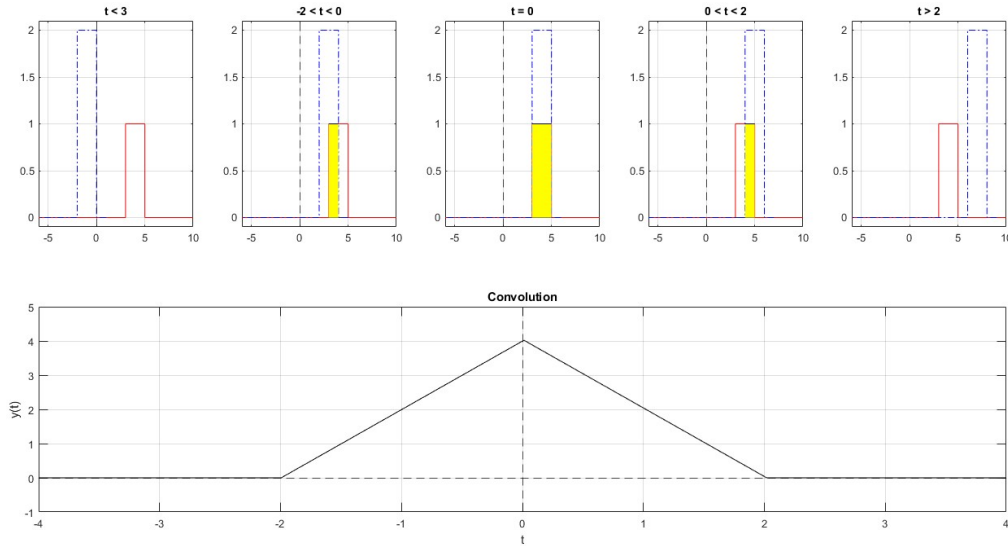
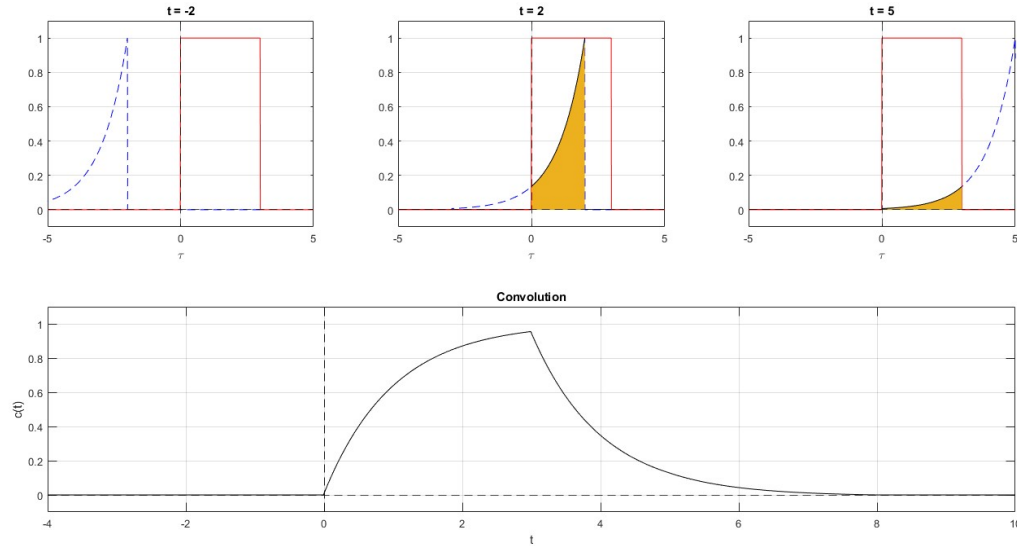


Figure 2:  $c(t)$  for each area of overlap/no overlap

As shown, the convolution  $c(t)$  as a function of time grows linearly from  $t = -2$  to  $t = 0$ , up to a value of  $(A \times B) * \text{width}$  (here, the widths of each rectangular pulse are equal), and then decreases linearly from  $t = 0$  to  $t = 2$ .

- f) For  $c(t) = e^{-t} * [u(t) - u(t-3)]$ , we may define an area of no overlap where  $t < 0$ , and an area of overlap for  $t > 0$ , which diminishes slowly as  $t \rightarrow \infty$ .

Fig. 3 shows several defined areas along with the convolution itself as a function of  $t$ .

Figure 3:  $c(t)$  for areas of overlap/no overlap

2.5.1 BIBO stability is also known as *external* stability, and is related to the stability of the system when provided with some external input. Given a bounded signal (ones that does not grow towards infinity over time), a BIBO-stable system provides a bounded output (one that does not grow towards infinity over time). As one might expect, BIBO stability can be determined from the system's impulse response  $h(t)$ , where BIBO-stable systems meet the criteria

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$

BIBO stability is a natural consequence of asymptotic or *internal* stability, which, as the name suggests, depends on the system's internal dynamics when presented with no input signal. A system's asymptotic stability can be determined based on its characteristic roots; if any lie in the right half of the  $s$ -plane, the system is asymptotically unstable.

- a) (i) To determine BIBO stability, we'll investigate the impulse response  $h(t)$  of the system described by  $(D^2 + 8D + 12)y(t) = (D - 1)x(t)$ . Using the simplified impulse matching method shown before and solving for  $h(t)$  in MATLAB,

$$h(t) = \left( \frac{e^{-6t} + 3e^{-2t}}{4} \right) u(t).$$

Then, evaluating the absolute integral of  $h(t)$ ,

$$\int_0^{\infty} |h(\tau)| d\tau \approx 0.244,$$

and so the system is **BIBO-stable**.

- (ii) BIBO-stability on its own does not imply asymptotic stability, so we'll check the criteria for asymptotic stability as well.

Solving the system's characteristic equation  $\lambda^2 + 8\lambda + 12 = 0$ , we will find characteristic modes

$$\lambda_1 = -6$$

$$\lambda_2 = -2$$

Since both resultant characteristic modes,  $e^{-6t}$  and  $e^{-2t}$ , have a negative real part, and are thus in the left side of the  $s$ -plane, the system is considered **asymptotically stable**.

- b) (i) Starting again with BIBO stability, we will derive the impulse response of  $D(D^2 + 3D + 2)y(t) = (D + 5)x(t)$  using MATLAB,

$$h(t) = (5e^{-2t} + \frac{5}{2})u(t).$$

Evaluating the integral,

$$\int_0^\infty |h(\tau)|d\tau = \infty,$$

and so the system is **BIBO-unstable**.

- (ii) Evaluating asymptotic stability, we may use the characteristic equation  $\lambda^3 + 3\lambda^2 + 2\lambda = 0$ ; finding its roots:

$$\begin{aligned}\lambda_1 &= 0 \\ \lambda_2 &= -2 \\ \lambda_3 &= -1,\end{aligned}$$

resulting in characteristic modes of 1,  $e^{-2t}$ , and  $e^{-t}$ . Since one of the characteristic roots is  $\lambda_1 = 0$ , which lies *on* the imaginary axis of the s-plane, we would consider the system to be **marginally stable**; that is, when presented with a small input or disturbance, it will eventually stabilize, but in a different state from its initial state prior to the input.

- c) (i) Once again, for BIBO stability, we will analyze the impulse response of  $D^2(D^2 + 2)y(t) = x(t)$ ,

$$h(t) = (\frac{t}{2} - \frac{\sqrt{2} \sin(\sqrt{2}t)}{4})u(t).$$

Taking the absolute integral of  $h(t)$ ,

$$\int_0^\infty |h(\tau)|d\tau = \infty,$$

and so the system is *BIBO-unstable*.

- (ii) Analyzing the characteristic roots for asymptotic stability,

$$\begin{aligned}\lambda_1, \lambda_2 &= 0 \\ \lambda_3 &= j\sqrt{2} \\ \lambda_4 &= -j\sqrt{2}.\end{aligned}$$

Since there are repeated roots on the imaginary axis, we consider this system **asymptotically unstable**, which matches our prior assessment of BIBO-instability.

- d) (i) For BIBO stability, we will first find the impulse response of the system  $(D + 1)(D^2 - 6D + 5)y(t) = (3D + 1)x(t)$ ,

$$h(t) = \frac{2}{3}e^{5t} - \frac{1}{2}e^t - \frac{1}{6}e^{-t}.$$

Taking its absolute integral,

$$\int_0^\infty |h(\tau)|d\tau = \infty,$$

and so the system is **BIBO-unstable**.

- (ii) Solving the system's characteristic equation results in the following roots:

$$\begin{aligned}\lambda_1 &= 5 \\ \lambda_2 &= -1 \\ \lambda_3 &= 1,\end{aligned}$$

and since two of the roots lie in the right side of the s-plane (positive real parts), the system is **asymptotically unstable**.